Yimo Liu

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Education

University of Florida Aug. 2025 – May 2029
Bachelor of Science in Data Science, Bachelor of Arts in Economics | GPA: 4.0/4.0 *Gainesville, FL*
Relevant Coursework: Intermediate Microeconomics, Regression Analysis, Data Structures & Algorithms
Certifications: Google Data Analytics Certificate, Coursera *May 2026*

Sickles High School Aug. 2022 – May 2025
ACCEL Standard Diploma *Tampa, FL*
AP Scholar with Distinction (2024, 2025): 13 AP exams, all scored 5
All-State Orchestra, Violin: Top 3 in graduating class (2023, 2024, 2025)

Research & Projects

NVIDIA Basketball Shot Analysis | NVIDIA Brev, Computer Vision, Sports Analytics Sep. 2026 – Present

- Lead the Data Science and Informatics project team
- Developing a GPU-accelerated pipeline utilizing NVIDIA Brev to analyze basketball jump shot quality
- Use computer vision to conduct player tracking, court-coordinate mapping, and shooter pose estimation to extract spatial context and body-mechanics features for jump shots
- Benchmarking tracking against SportsMOT ground truth and comparing context-only and pose-augmented expected-make models

Public University Access & Household Children | R, Multiple Linear Regression Sep. 2026 – Present

- Lead the University Economics Society project team
- Analyzing 2024 American Community Survey Public Use Microdata Sample data to examine the relationship between household distance to a public university and number of children
- Investigating geographic and household-level patterns relevant to higher-education access and family outcomes

Assisted Housing Inventory EDA | Python Mar. 2026 – May 2026

- Lead the UF Statistics Club project team
- Conducting exploratory data analysis on assisted housing inventory in Alachua County, analyzing 9,200+ data points to identify trends and socioeconomic relationships
- Presented Spring 2026 findings to club members on Apr. 22, 2026

ASA DataFest 2026 | *Stormont Vail Health Length-of-Stay Prediction* | Python, scikit-learn, pandas Apr. 2026

- Led 5-person team over 48 hours, predicting length of stay from 7.6M patient encounters across 230M+ total data points
- Designed 20-feature modeling approach including Journey Depth metric to capture diagnosis recurrence; applied log transformation to address heavy right-skew in target variable
- Compared linear regression, random forest, and gradient boosting models; gradient boosting achieved $R^2 = 0.58$ and MAE of 46.6 hours, outperforming linear baseline
- Identified admission status as the single greatest LOS determinant via feature importance analysis; presented findings and methodological limitations to statistics faculty and grad student panel

Federal Reserve and Inflation Control | *Calculus, Python, EDA, Regression* Aug. 2025 – Dec. 2025

- Lead authored 16-page paper with team of 3 evaluating Federal Reserve effectiveness at controlling U.S. inflation
- Analyzed 15+ years of Fed Funds Rate and CPI data with 2,600+ data points via exploratory data analysis and inferential statistics
- Designed custom responsiveness metric utilizing weighted linear regression and multi-variable calculus

Leadership & Experience

Undergraduate Researcher Aug. 2026 – Present
UF Virtual Learning Lab, University of Florida *Gainesville, FL*
- Building dataset for Storiza, an AI-powered K-2 literacy tool, covering 86 South Carolina school districts, containing AI policy, competitor presence, device-accessibility, literacy scores, and financial data
- Conducting procurement research on 8 competing vendors, finding contracts and intent-to-buy documents by examining records from the Hillsborough County library system and school districts
- Gathered 1,459 unique current and past third-party vendor contracts from the Hillsborough County library system through public-records requests, including contract values, dates, and renewability

Teaching Assistant / Liaison Officer Aug. 2026 – Dec. 2026
ECO3101 Intermediate Microeconomics, University of Florida *Gainesville, FL*
- Serve as liaison between the student body and instructor, managing attendance and communication
- Help define homework grading criteria and set up the Applied Economic Analysis project, where students present real-world applications of course material

Research Director Apr. 2026 – Present
Data Science and Informatics, University of Florida *Gainesville, FL*
- Placed 5 students in UF research labs by hosting research-access events and fostering faculty-student connections
- Lead student research initiatives as primary liaison between DSI's 2000+ members and UF research labs
- Developing and maintaining the UF Research Guide from scratch, connecting DSI students to on-campus research groups through up-to-date lab openings, faculty contacts, and application strategies

Networking Coordinator Sep. 2025 – Apr. 2026
Data Science and Informatics, University of Florida *Gainesville, FL*
- Transitioned DSI's Yearly Research Symposium from hybrid to fully in-person format, drawing 100+ student and professional attendees and showcasing 10+ research posters and projects
- Reached 50+ companies, 25+ research labs, and 20 speaker candidates, securing 6 speakers for the Symposium
- Secured $8,000+ in funding through corporate and academic sponsorships

VP of Career Development Apr. 2026 – Present
University Economics Society, University of Florida *Gainesville, FL*
- Invite professors, PhD students, and research labs to present economics research and career paths to UES members
- Organize flagship events: Undergrad Research Symposium (peer econ research presentations) and Career Panel (faculty and industry Q&A)
- Quantitatively analyze weekly economic topics and present research findings at each club meeting

Project Director Mar. 2026 – May. 2026
UF Statistics Club, University of Florida *Gainesville, FL*
- Led a semester-long research project on Assisted Housing Inventory, directing project agenda and coordinating team members

Hackathons

SASEHacks 2026 | **Best Finance Project** | *CartCoach* | Python, TypeScript, Google Gemini API Mar. 2026
- Designed and built Chrome Extension to identify and reduce impulse buying and build financial habits
- Developed FastAPI/MongoDB backend storing spending history, powering monthly ledger and purchase tracking
- Built dashboard with spending breakdowns, savings goal tracking, and Gemini-powered financial assistant
- Led a 4-person team over 24 hours

SwampHacks XI | *RePlot* | Python, FastAPI, React, Google Gemini, Google Maps, Mapbox Jan. 2026
- Built backend and led presentation for urban infill development tool analyzing land for redevelopment potential
- Designed 100-point scoring system across ROI, physical traits, regulations, and community impact
- Integrated Google Gemini for per-parcel analysis with logic to exclude public amenities; incorporated IPCC Carbon Accounting Standards for environmental impact metrics

Gator Hacks IV | **Overall 3rd Place** | *TravelBuddy AI* | Python, Gemini, Vision AI, Places API Oct. 2025
- Built backend LLM agent using Google Gemini API to generate travel and art recommendations
- Improved LLM response speed by 12% through prompt engineering and A/B testing
- Led 3-person team over 48 hours in project planning, API integration, and presentation

ShellHacks 2025 | *Clinix* | Python, FastAPI, React, Google Gemini, Tailwind CSS Sep. 2025
- Developed AI-powered hospital front-desk assistant for appointment scheduling, medical records retrieval, and insurance cost estimation
- Designed privacy-first architecture where patient data is never exposed to the LLM
- Integrated backend with Google Sheets, Calendar, and web retrieval; deployed on Vercel and Render
- Led 3-person team over 36 hours

Technical Skills

Programming Languages: Python, R, C++, Java, SQL
Frameworks & Libraries: FastAPI, MongoDB, NumPy, pandas, scikit-learn, SciPy, Matplotlib
Developer Tools: Git, Google Cloud Platform, Jupyter Notebook